DAY 18 — AI ETHICS & LIMITATIONS

Ethics, Ethical Theories, Fairness, Bias, and AI Limitations

Learning outcome:

- Identify AI limitations and hallucinations,
- Apply ethical thinking,
- Recognize fairness risks
- Evaluate simple cases responsibly.

Hour 1: Why ethics matters in AI

Theory:

- AI ethics is the study and practice of using AI in ways that respect people, reduce harm, and support fair outcomes.
- AI systems can affect opportunities, privacy, safety, money, reputation, and access to services.
- Ethical thinking is necessary before, during, and after an AI system is used.

Why risks arise:

- Training data can be incomplete or biased.
- The system may make errors or hallucinate facts.
- People may trust AI output too much.
- A tool can be used for a purpose it was not designed for.
- Important decisions may be automated without proper review.

Key ethics questions:

- Who may benefit?
- Who may be harmed or excluded?
- What personal or sensitive data is involved?
- What happens if the AI is wrong?
- Who checks the result and takes responsibility?

Stakeholders:

People or groups affected by an AI decision, such as learners, job applicants, customers, employees, managers, and the public.

Example:

An AI tool summarizes student complaints. It can save time, but a human must review urgent complaints so serious cases are not missed.

Hour 1: Easy practice — stakeholders & harm

Scenario:

“A training center uses AI to sort learner support messages into ‘urgent’ and ‘normal.’”

Tasks:

1. Name three stakeholders.
2. Name one benefit of the tool.
3. Name two possible harms if it is wrong.
4. Suggest one human-check step.

Write short answers—one line each.

Example answers:

- Stakeholders: learners, support staff, training manager.
- Benefit: staff can respond faster.
- Harm: an urgent message may be classified as normal; learner data may be exposed.
- Human check: support staff review all messages marked urgent and sample normal messages.

Easy AI prompt:

“For this AI support-message system, list stakeholders, one benefit, two risks, and one human oversight step. Use simple language and do not invent facts.”

Hour 2: Introduction to ethical theories

Theory:

Ethical theories are structured ways to think about whether an action is right or wrong. They help people discuss difficult choices instead of relying only on personal opinion.

Utilitarianism:

- Choose the action expected to create the greatest overall benefit and least overall harm.
- Question: “Which option produces the best total outcome for the most people?”

Deontology:

- Focus on duties, rules, rights, and respect for persons.
- Question: “Does this action follow our duties and respect people’s rights?”

Virtue ethics:

- Focus on the character and values shown by the decision-maker.
- Values include honesty, fairness, courage, care, and responsibility.
- Question: “What would a responsible and fair person do?”

Important note:

No single theory solves every problem. Use multiple viewpoints, explain trade-offs, and involve people with responsibility for the decision.

Example:

Using private learner data without permission might produce a useful service, but it can violate privacy rights and fail the tests of duty and trustworthiness.

Hour 2: Easy practice — three ethical lenses

Scenario:

“An organization wants to use AI to monitor employee emails for signs of stress.”

Answer one simple question for each lens:

- Utilitarianism: What possible benefit and harm should be compared?
- Deontology: Which right or duty matters most?
- Virtue ethics: Which value should guide the organization?

You do not need a perfect answer—state your reasoning clearly.

Example responses:

- Utilitarianism: benefit may be early support; harm may be privacy loss and fear.
- Deontology: respect privacy and obtain clear consent.
- Virtue ethics: act with care and honesty, not secret surveillance.

Easy AI prompt:

“Analyze this AI email-monitoring proposal using utilitarianism, deontology, and virtue ethics. Give one simple question and one concern for each theory. Avoid a final legal conclusion.”

Theory:

- Ethical AI decisions require facts, context, affected people, and a process for reviewing harm.
- The same AI tool can be low-risk in one context and high-risk in another.
- A spelling assistant is lower risk than an AI system used to screen job applicants or approve loans.

Simple ethical decision process:

1. Define the purpose and expected benefit.
2. Identify affected people and possible harms.
3. Check data, privacy, bias, and accuracy risks.
4. Consider alternatives and safeguards.
5. Decide who reviews, approves, and monitors the system.

Human oversight:

- Human-in-the-loop: a person reviews each important decision.
- Human-on-the-loop: a person monitors the system and can intervene.
- Human-out-of-the-loop: no meaningful human review; use with extreme caution for high-impact tasks.

Example:

AI can draft a list of interview questions, but a hiring manager should review them for relevance, fairness, and job fit.

Good documentation:

Write down the purpose, data source, limitations, owner, review date, and how people can report a problem.

Hour 3: Easy practice — AI hiring support

Scenario:

“A company uses AI to summarize CVs and rank candidates for interview.”

Tasks:

1. Identify one possible benefit.
2. Identify two possible ethical risks.
3. Choose a human-oversight method.
4. Write one safeguard to reduce bias.

Use job-related skills only; do not use age, gender, religion, name, photo, or other irrelevant personal information.

Example response:

- Benefit: saves time organizing applications.
- Risks: the ranking can reflect bias; a qualified candidate may be wrongly excluded.
- Oversight: human-in-the-loop; recruiter reviews every shortlist decision.
- Safeguard: compare candidates using the same skills-based scorecard.

Easy AI prompt:

“Create a simple ethical checklist for using AI to summarize CVs. Include fairness, privacy, accuracy, human review, and documentation. Keep it to five checks.”

Hour 4: Fairness, bias & AI limitations

Theory: fairness and bias

- Fairness means people should not receive unjustly different treatment because of irrelevant personal characteristics.
- Bias is a systematic pattern that can create unfair results.
- Bias can enter through data, labels, design choices, prompts, users, or how results are applied.

Common bias sources:

- Historical data reflects past unequal treatment.
- Some groups are missing or underrepresented in the data.
- A proxy variable indirectly represents a sensitive trait.
- Human reviewers interpret AI output unfairly.

Theory: AI limitations

- AI can generate plausible but false statements, often called hallucinations.
- AI may be outdated, incomplete, inconsistent, or unable to access reliable sources.
- AI does not automatically understand local context, policy, or human values.
- Fluent language is not proof of accuracy.

Verification steps:

- Check important claims against trusted source material.
- Verify numbers with calculations or original data.
- Ask for uncertainty, assumptions, and alternative explanations.
- Do not use AI output as evidence unless the sources are checked.

Hour 4: Easy practice — catch a risky answer

AI output:

“Candidate A is definitely the best person because they have the most professional-sounding name and are likely to fit our culture.”

Tasks:

1. Identify two problems with this output.
2. Rewrite the decision rule using fair, job-related criteria.
3. State one fact a human must verify.

Hint: A person’s name is not evidence of job ability.

Example response:

- Problems: irrelevant personal characteristic; overconfident claim without evidence.
- Fair rule: compare verified skills, relevant experience, and structured interview scores.
- Verify: each candidate’s actual work experience and skills evidence.

Easy AI prompt:

“Rewrite this hiring recommendation so it uses only job-related criteria. Flag any unsupported or biased reasoning. Use cautious language and recommend human review.”

Hour 5: Tools & simple metrics for fairness

Theory:

- Fairness evaluation asks whether an AI system performs differently for different groups in ways that may be unjust.
- Metrics help identify a possible issue; they do not prove fairness by themselves.
- Always interpret results with context, sample size, data quality, and human impact.

Simple evaluation questions:

- Are all relevant groups represented in the data?
- Does the system approve, reject, or flag groups at very different rates?
- Does it make more errors for one group?
- Are decision criteria relevant to the purpose?
- Can people challenge or correct a decision?

Useful basic measures:

- Selection rate: selected people / total people in a group.
- Error rate: incorrect results / total results for a group.
- Representation: whether groups are present in sufficient numbers.

Example:

If 8 of 20 Group A applicants and 2 of 20 Group B applicants are selected, selection rates are 40% and 10%. This difference requires investigation; it does not automatically prove discrimination.

AI can calculate and summarize, but people must decide what is fair and what action is needed.

Hour 5: Easy practice — compare selection rates

Simple data:

- Group A: 20 applicants; 8 selected
- Group B: 20 applicants; 2 selected

Tasks:

1. Calculate each selection rate.
2. Describe the difference without making an accusation.
3. List two questions to investigate.

Use respectful language: “This result needs review,” not “The system is proven unfair.”

Expected answer:

- Group A selection rate = 8 / 20 = 40%
- Group B selection rate = 2 / 20 = 10%
- The selection rates differ and should be investigated.
- Questions: Were both groups assessed with the same job criteria? Is the data complete and accurate?

Easy AI prompt:

“Calculate selection rates for two groups and explain the difference carefully. List two investigation questions. Do not claim discrimination without additional evidence.”

Hour 6: Case analysis & responsible action

Theory:

- A case study helps trainees apply ethics to a realistic situation.
- A strong response identifies the purpose, stakeholders, evidence, risks, safeguards, and accountable owner.

Case-analysis template:

1. What is the AI system trying to do?
2. Who is affected?
3. What can go wrong?
4. Which ethical principles apply?
5. What safeguard should be added?
6. Who approves and monitors the system?

Mitigation options: improve data, remove irrelevant features, add human review, test results by group, provide appeal routes, limit use, or stop deployment.

Responsible action:

- Do not ignore an ethical concern because the tool is convenient.
- Escalate high-impact, unfair, unsafe, or privacy-sensitive issues.
- Keep a record of decisions, risks, safeguards, and review dates.
- Update the process when evidence shows a problem.

Key message:

Ethical AI is not only about the technology. It is about people, context, accountability, and continual review.

Hour 6: Easy practice — group case & exit ticket

Group case:

"An AI chatbot gives financial guidance to students. It sometimes gives confident answers without showing sources."

Group tasks:

1. Name two possible harms.
2. Name one AI limitation shown in the case.
3. Suggest two safeguards.
4. Identify who should review the chatbot's advice.

Present your answer in four short bullets.

Example response:

- Harms: students may make harmful money choices; false information may spread.
- Limitation: hallucinated or unsupported information.
- Safeguards: show approved sources; direct users to a qualified human adviser for high-impact decisions.
- Reviewer: trained financial-support staff or an approved expert.

Exit ticket:

- One AI limitation I will check: _____
- One fairness question I will ask: _____
- One decision needing human review: _____

Course alignment: Day 18 from the supplied module—importance of AI ethics, utilitarianism/deontology/virtue ethics, application in AI contexts, fairness, fairness evaluation, and biased-AI case studies.